

It is very important for people to find out if they are buying the right S/D station as there are a few available in the market that have microcontroller but are not smart stations with sensors. Making a wrong selection may leave one with a station that would deteriorate over time and would not be of any use further. —Kanishka Shah, branch manager, Entesla

to be curious about what this term is and why it is important. To clarify this term, let me take an example.

We set our S/D stations to a specific temperature for efficient work. However, do we ever bother to know if the tip of the iron has reached that temperature, or how close is it to the set temperature? The temperature accuracy gives us that idea. Ideally, the accuracy should be $\pm 0.001^\circ$, which is impossible, but an S/D with $\pm 5^\circ$ of temperature accuracy is considered suitable.

An S/D station with adjustable irons and tips should have the accuracy calibration option to maintain the working efficiency of the S/D station.

Tip-to-ground ratings. This rating refers to two different parameters: potential and resistance. Tip-to-ground potential is the existing voltage between the tip of the S/D iron and the common ground of the station, and the tip-to-ground resistance is the resistance between the two.

The first is the leakage potential, which should not be more than 2mV, while the resistance should not be more than 5 Ω . Otherwise, the overall current will leak out and the station will never provide the required efficiency at work.

New features to look for

You can never make the best choice if you do not have information on what is new in the market. Although you will find a few new styles of S/D stations with respect to their looks and design, the ones listed below are some of the new features.

Ultra-fast thermal recovery. You might have got a synopsis of thermal recovery in the previous few paragraphs, but ultra-fast thermal recovery is something even beyond that. Only a soldering station with high wattage like

120W or 150W can provide this type of recovery, allowing you to operate it at a very low temperature set point. This minimises your concern about the influence of heat on the components and the tip of the iron.

Such soldering stations have an auto-sleep function along with a power shut-off that gets activated once the set temperature is reached, preventing the soldering iron from getting over-heated and thus increasing life of the station and the tools attached. The auto-sleep also prevents oxidation of the tip of the iron.

These soldering stations are used for fine SMD soldering, high mass soldering purposes and soldering parts of a heavy automotive.

Temperature pre-set. All the new S/D stations that have come up recently have the temperature pre-set option, where three frequently used temperatures can be set in advance and then the required temperature can be recalled with just a single push of a button. This is an added advantage to the temperature control function that we usually look for in any S/D station.

Ultra-light soldering iron. A light soldering iron is advisable for comfort at work. Today, you can get a soldering station with an ultra-light-weight soldering iron that weighs not more than 95 grams. The plug-in type soldering tips with different shapes and sizes further add to this feature by providing the optimum working facility for compact boards with more than hundreds

of components to be soldered in.

User-friendly. You can also see simple and user-friendly S/D stations that not only provide different factory settings for adjustment of variable temperature, calibration value and standby time, but also provide the ability to use fixed temperatures, interlock and energy levels with the use of PC software or a micro SD card. Another special feature in such type of S/D stations is the shut-down and standby function that helps save energy, increasing the life of the iron tip at the same time.

Thermal control and lock. Have you ever thought of controlling the heat of the S/D station and locking it at a certain point so that the iron does not get over-heated, eventually damaging the project? Think now, if you haven't before.

The new S/D stations available now come with a control card that turn off power to the station if the iron tip reaches the set temperature. Now, to check if the tip of the iron has reached the set point or not, they have temperature sensors incorporated at the tip.

Such S/D stations provide certain features that are required for industrial manufacturing. These features are: compact design for energy saving, information regarding the errors and offset functions to moderate the heat. Such S/D stations are considered suitable for a wide range of applications starting from few touch-ups to heavy-duty work.

Temperature sensors and alarm. This is a soldering station that allows no fluctuation in temperature during continuous operation at heavy load. To support this feature, such stations have high-sensitivity sensors connected to an alarm that turns on when the temperature of the tip goes above or below the maximum and the minimum temperature settings. The alarm again turns on if the iron is left in contact with the component or the board longer than the

There are some soldering stations where temperature governs the power delivery. The idea behind it is that, with the rate of change of temperature of the station, the joint temperature should reach the activation temperature. —One of the vendors from the soldering industry